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# A Review of Electric Vehicle Charging Systems: Converter Architectures, Control Optimization, and PV-Integrated Energy Management

1st Mubaarak A.A. Saif
*Department of Energy Systems Engineering*
*Faculty of Technology*
*Kocaeli University*
Kocaeli, Turkiye
ORCID:0000-0002-5120-6792
mubaarak.saif@kocaeli.edu.tr

2nd Salah F. S. Saeed*
*Faculty of Engineering*
*and Information Technology*
*Taiz University*
Taiz, Yemen
ORCID: 0009-0000-9485-7467
salahfaisal589@gmail.com

3rd Abdullah Haidar
*Faculty of Science and Engineering*
*Swansea University*
Swansea, UK
ORCID: 0009-0003-5344-7682
843455@swansea.ac.uk

4th Alzahraa F. M. Almakhathy
*Ne'marassam School*
Taiz, Yemen
ORCID: 0009-0003-2238-7509
alzahra@ne3marasam.org

5th Malak M. N. Al-Koshab
*Ne'marassam School*
Taiz, Yemen
ORCID: 0009-0003-2911-0289
Malak@ne3marasam.org

6thNora H. A. Qasem
*Ne'marassam School*
Taiz, Yemen
ORCID: 0009-0001-6993-1052
noraqasem@ne3marasam.org

***Abstract*—Electric vehicle (EV) charging research has evolved rapidly from converter-level hardware innovation to intelligent control design and, more recently, to renewable-integrated charging and station-level energy management. However, the literature remains fragmented when charger topology, control strategy, and system integration are treated as separate themes. This review addresses that fragmentation through a layered analytical framework that organizes the field into three interconnected research streams: converter and charger architecture studies, control and optimization studies, and integrated PV–EV charging and energy-management studies. The review shows that architectural research has primarily advanced voltage adaptability, bidirectional operation, efficiency, and ripple mitigation, while control-oriented studies have improved DC-link regulation, current tracking, harmonic suppression, and dynamic response. At the system level, PV-assisted and grid-interactive charging studies have expanded the scope of EV charging toward coordinated energy routing, renewable utilization, and supervisory operation. Despite these advances, several persistent limitations remain, including narrow optimization objectives, limited full-scale validation, weak incorporation of battery ageing and electrothermal effects, insufficient operational realism, and the absence of unified cross-layer co-design frameworks. The synthesis suggests that the next stage of progress will depend on hierarchical and reproducible system-level design approaches that jointly integrate converter design, controller synthesis, battery-aware management, renewable coordination, and grid-responsive functionality. In this sense, future EV charging systems must be developed not merely as efficient chargers, but as scalable, adaptive, and deployment-ready energy platforms.**

***Index Terms*—Electric Vehicle Charging Systems; Converter Architectures; Bidirectional Chargers; Control Optimization; PV-Integrated Charging; Energy Management; Vehicle-to-Grid; Power Quality**

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*Corresponding author:Salah F. S. Saeed

## I. Introduction

The rapid growth of electric vehicles (EVs) is transforming the role of charging infrastructure from a peripheral support technology into a central component of future power and transportation systems. Modern EV charging systems are no longer expected only to deliver energy efficiently; they are increasingly required to operate under wide battery-voltage ranges, support fast charging, enable bidirectional power exchange, coordinate with renewable generation, and interact intelligently with the grid [1]–[7]. As a result, EV charging has emerged as a multidisciplinary research field shaped simultaneously by power electronics, control engineering, battery management, renewable integration, and energy-system operation.

This expansion has produced a large and technically diverse body of literature. At the hardware level, many studies focus on converter topology, charger architecture, gain enhancement, soft switching, interleaving, isolation, and bidirectional operation [8]–[14]. A second stream emphasizes control refinement through predictive, adaptive, nonlinear, and optimization-assisted strategies intended to improve current tracking, harmonic performance, transient response, and DC-link regulation [15]–[21]. More recently, a third stream has moved toward integrated PV–EV charging, grid-support services, and station-level supervisory energy management, where EV charging is treated as part of a coupled renewable-aware and grid-interactive energy environment [22]–[29]. Although these streams collectively reflect substantial technical progress, they are often examined in isolation, which obscures their deeper interdependence.

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This fragmentation is increasingly problematic because charger architecture, control synthesis, battery-side regulation, renewable coordination, and station-level operation are not independent design layers. Decisions made at the converter level shape the controllability and operating envelope of the charger; control choices affect power quality, dynamic stability, and battery stress; and renewable integration or supervisory management can introduce additional coupling through common DC links, shared storage, and grid-side constraints [3], [4], [30], [31]. Consequently, a review that only compares topologies, or only benchmarks controllers, cannot fully explain how the field is evolving or where its most important unresolved challenges lie.

To address this issue, the present review adopts a structured taxonomy that organizes the literature into three complementary research layers: converter and charger architecture studies, control and optimization studies for EV charging systems, and integrated PV–EV charging and energy-management studies, as illustrated in Fig. 1. This framework is not intended merely as a classificatory device; rather, it serves as an interpretive structure for understanding the layered development of EV charging research from hardware realization, to control-oriented refinement, and ultimately to coordinated renewable-integrated charging ecosystems. By analyzing the literature through this layered lens, the review clarifies the dominant technical contributions of each research stream, the limitations of studying each layer independently, and the growing importance of cross-layer coordination.

On this basis, the review makes three main contributions. First, it synthesizes the recent EV charging literature within a coherent analytical framework that links topology, control, and system integration rather than treating them as disconnected themes. Second, it identifies the principal cross-group limitations that continue to constrain the field, including narrow optimization objectives, limited validation maturity, weak battery-aware design, insufficient representation of stochastic operating conditions, and uneven reporting practices [2], [32]–[34]. Third, it outlines future directions for next-generation EV charging systems, emphasizing hierarchical co-design, transparent benchmarking, realistic validation, and scalable architectures capable of integrating charging efficiency, renewable coordination, battery-conscious management, and grid-responsive operation within a unified platform.

## II. Taxonomy and Analytical Framework

The literature on electric vehicle charging systems has expanded across multiple technical directions, yet its development remains analytically fragmented when converter design, control strategy, and system-level integration are treated as isolated themes. To address this fragmentation, this review adopts a structured taxonomy that organizes the reviewed studies into three complementary research layers: converter and charger architecture studies, control and optimization studies for EV charging systems, and integrated PV–EV charging and energy-management studies, As shown in Fig. 1.

It should be noted that this article is designed as a thematic and structured literature synthesis rather than a protocol-driven systematic review. Therefore, the adopted framework is not intended to represent a PRISMA-based screening protocol, bibliometric study, or quantitative meta-analysis. Instead, it provides an analytical taxonomy for organizing representative studies across three major research layers: converter and charger architectures, control and optimization strategies, and integrated PV–EV charging and energy-management systems. The reported trends are therefore interpreted as synthesis-based patterns derived from comparative analysis rather than as statistically pooled effect-size estimates. To reduce selection imbalance, the reviewed studies were organized according to these predefined layers rather than according to a single algorithmic family or converter type, ensuring that hardware-level, control-level, and system-level contributions were each represented as distinct but interdependent research streams.

This three-layer framework is not intended merely as a classificatory tool, but as an interpretive structure that reflects the technical evolution of the field—from hardware realization and power-conversion capability, to control intelligence and performance refinement, and ultimately to coordinated, renewable-integrated charging ecosystems. This analytical framing enables a more coherent reading of the literature by clarifying the dominant contribution of each group, the limitations of examining each layer in isolation, and the increasing need for cross-layer integration in next-generation EV charging research.

### A. Group I: Converter and Charger Architecture Studies

The studies reviewed in this stream demonstrate that the early development of EV charging research has been shaped primarily by converter- and charger-level architectural innovation. As synthesized in Table I, the literature does not converge around a single dominant topology; rather, it reflects a sequence of design responses to recurring engineering demands, including higher efficiency, wider voltage adaptability, reduced ripple, bidirectional power transfer, and improved charging flexibility. Accordingly, the architectural layer should be understood not merely as a hardware implementation choice, but as the foundational design domain that determines the operational envelope within which all subsequent control and energy-management strategies must function.

A prominent direction within this body of work is the development of resonant and isolated bidirectional structures intended to combine high conversion efficiency with broader charging capability. In [8], a single-stage current-fed resonant AC–DC charger integrates power-factor correction and soft-switching operation for 400 V and 800 V battery platforms, reporting a peak efficiency of 96.48%, a power factor approaching unity, and THD of 2.8%. A related bidirectional extension is presented in [9], where a three-phase voltage-source converter and a soft-switching DAB stage are combined to support G2V/V2G operation in an on-board charger configuration. Similarly, [10] adopts a more elaborate isolated two-stage structure built around a quadratic boost converter,

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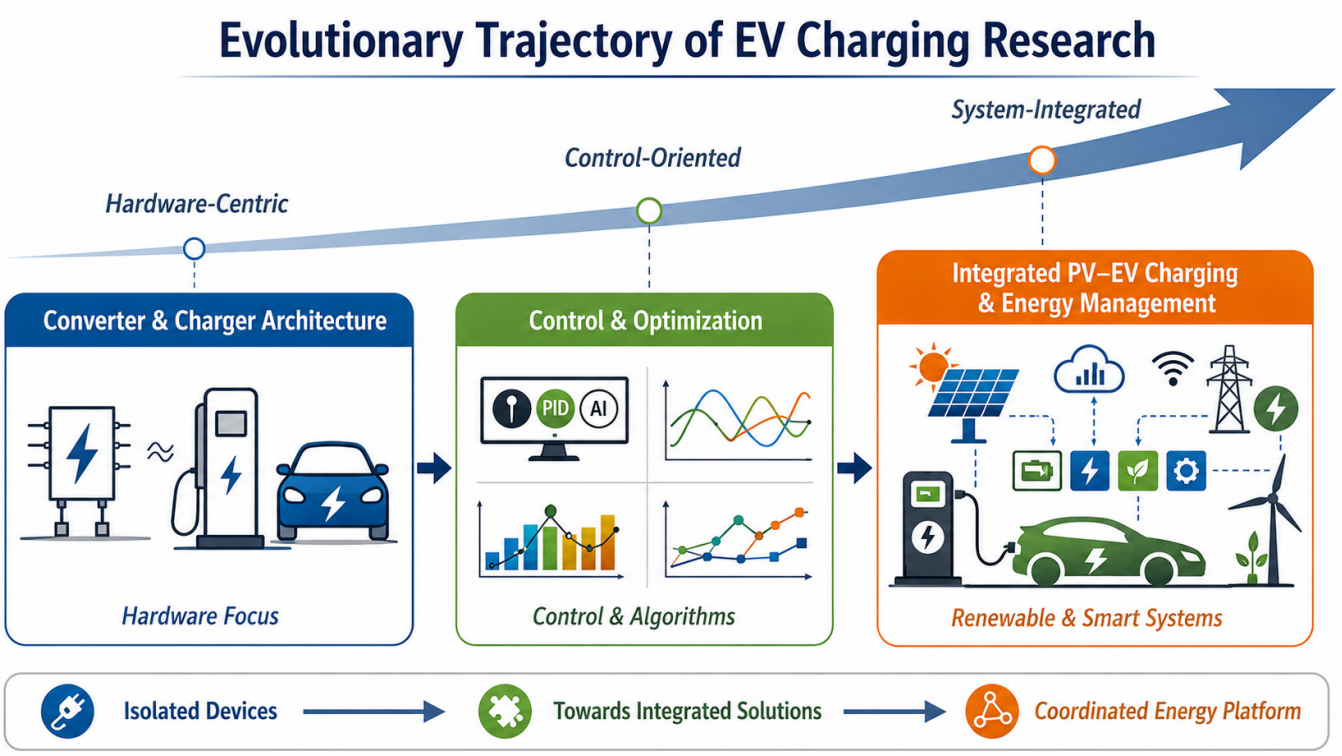



Fig. 1. Evolutionary trajectory of EV charging research, illustrating the field's progression from hardware-centric studies on converter and charger architecture, through control- and optimization-oriented investigations, to system-integrated research on PV–EV charging and energy-management platforms.

DAB converter, and T-type inverter, thereby extending the conversion capability for FCBEV applications while maintaining THD below 2.35% and a DC-link voltage of approximately 700 V. Taken together, these studies indicate that resonant and isolated architectures remain central when the design objective is to reconcile high efficiency, bidirectional operation, and wide voltage compatibility. However, Table I also makes clear that these gains are frequently supported more strongly by simulation than by experimental validation, which limits the immediate translational strength of several proposed designs.

A second major trajectory is the use of interleaved and multiphase structures as a hardware-level means of improving current quality, stress distribution, and load-range efficiency. This tendency is most clearly represented by [12], where interleaving, intraleaving, and phase shedding are unified within a multiphase bidirectional buck–boost architecture to reduce losses across different operating conditions. The reported peak efficiency of 98.8–98.9% and improved light-load efficiency indicate that such architectures are valuable not only under rated operation, but also in preserving performance under variable loading. In a related manner, [13] uses phase-shifted interleaving in a high-gain non-isolated boost converter to reduce ripple and distribute current stress, achieving a peak efficiency of about 95.6% and a voltage gain of approximately four. These results confirm that interleaving has become more than a secondary implementation feature; it is now a principal architectural mechanism for shaping ripple behavior, current sharing, and efficiency stability. Yet, as the table also shows, these benefits are not cost-free, since higher RMS currents, increased conduction losses, thermal sensitivity, and limited scalability continue to constrain broader deployment at higher power levels.

A third direction is the search for architectures that achieve substantial voltage gain or charging flexibility with reduced active-switch complexity. The modified SEPIC-based bidirectional charger in [11] illustrates this tendency through a wide-range interface designed for low-voltage batteries, prioritizing adaptability rather than high-power scaling. Its reported efficiency of 90.18–92.27% and ripple below 10% suggest practical relevance for lower-power applications, although its 1.1 kW validation scale limits the generality of its implications. A more gain-oriented example is provided in [14], which proposes a transformerless ultra-high-gain single-switch buck–boost converter with reduced switch stress and continuous input current. While the reported maximum efficiency of 91.4% and output voltage near 720 V confirm the feasibility of the approach, the associated dependence on a high passive-component count and open-loop operation highlights the practical limitations that often accompany structurally simplified high-gain designs.

### *B. Group II: Control and Optimization Studies for EV Charging Systems*

The literature examined in this stream reflects a clear shift in emphasis from converter realization to converter regulation.

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TABLE I
COMPRESSED COMPARISON OF CONVERTER AND CHARGER ARCHITECTURE STUDIES

| Ref. | Year | Topology | Contribution | Validation | Performance | Limitation |
|---|---|---|---|---|---|---|
| [8] | 2025 | Single-stage CF resonant AC–DC charger | 400/800 V resonant fast charger with integrated PFC + soft switching | Sim. + HIL | $(\eta_{pk})$ 96.48%; PF 0.998–0.999; THD 2.8% | No prototype; 800 V only simulated |
| [9] | 2024 | $3\phi$ VSC + SS-DAB | Soft-switching bidirectional OBC for G2V/V2G | Sim. | Battery $(\eta) > 95\%$ | No experiments; optimization not explicitly supported |
| [10] | 2025 | Quadratic boost + DAB + T-type inverter | Isolated two-stage bidirectional FCBEV charger | Sim. | THD $< 2.35\%$; $(V_{dc}) \approx$ 700 V | No hardware; incomplete GA reporting |
| [11] | 2025 | VSC + mod. SEPIC | Wide-range bidirectional charger for low-voltage batteries | Sim. + exp. | $(\eta)$ 90.18–92.27%; ripple $< 10\%$ | Low-power only (1.1 kW); no formal optimization |
| [12] | 2025 | Multiphase interleaved bidir. buck–boost | Interleaving + intraleaving + phase shedding for loss minimization | Sim. + exp. | $(\eta_{pk})$ 98.8–98.9%; light-load $(\eta)$ 97–99% | Higher conduction/RMS losses; not ideal at high power |
| [13] | 2024 | Interleaved high-gain non-isolated boost | Ripple/stress reduction via phase-shifted interleaving | Sim. + exp. | $(\eta_{pk})$ 95.6%; gain $\approx 4$; low ripple | Thermal/parasitic sensitivity; limited scalability |
| [14] | 2024 | Transformerless UHG single-switch buck–boost | Ultra-high-gain reduced-stress single-switch design | Sim. + exp. | $(\eta_{max})$ 91.4%; $(V_o) \approx$ 720 V | High passive count; open-loop only |

Once charger architectures became sufficiently capable of supporting bidirectional power transfer, acceptable efficiency, and wider operating ranges, the central challenge was no longer the hardware structure itself, but the extent to which its behavior could be controlled, refined, and stabilized under dynamic operating conditions. As reflected in Tables II, this body of work is therefore primarily concerned with improving DC-link regulation, current tracking, harmonic performance, transient response, and coordinated G2V/V2G operation through increasingly sophisticated control and optimization strategies.

A useful analytical distinction within this literature lies between studies that improve charger behavior through a reformulation of the control strategy itself and those that retain familiar controller families but strengthen them through optimization-assisted tuning. The first tendency is most clearly illustrated by [16] and [18]. In [16], the proposed AMP-DPC/ADPC framework enhances bidirectional charger performance across G2V, V2G, and V2L modes by combining adaptive predictive and direct power control. Its experimental results are particularly compelling, with THD reduced from 29.301% to 2.008% and settling time improved to 5–8 ms. The significance of this study lies not only in the reported numerical gains, but also in the fact that the improvement is achieved through a more suitable control law rather than through post hoc gain adjustment. A comparable shift in emphasis appears in [18], where a coordinated robust nonlinear control framework, supported by sliding-mode concepts and offline PSO tuning, improves tracking quality and reduces DC-bus error under bidirectional BEV charger operation. These studies suggest that part of the literature has already moved beyond the view of charger regulation as a mere tuning exercise and has begun to treat it as a deeper question of control-structure adequacy under coupled, nonlinear, and bidirectional operating conditions.

Even so, the dominant pattern in the reviewed corpus remains the use of optimization-assisted tuning for PI- and PID-based controllers. This tendency is especially clear in [15], [17], [19], [20], and [21]. In [15], Grey Wolf Optimization is used to tune PID gains in a grid-connected PV inverter, reducing THD to 3.7% while improving settling time and maintaining a power factor close to unity. In [17], the AIChOa framework is applied to PI tuning in a bidirectional off-board charger, with the reported benefits expressed through better convergence and lower error measures. A more strongly dynamic formulation is presented in [19], where WHPSO tunes cascaded PI controllers for DC-link and battery-current regulation, reducing overshoot from 27.5% to 3.75% and shortening settling time from 125 ms to approximately 20–25 ms. In [20], SSA is used to optimize PI gains associated with DC-link regulation, MPPT support, and grid-current quality, yielding THD reduction to 2.50% and faster settling under irradiance variation. The most coordinated tuning contribution in this subset appears in [21], where ABC is used for system-wide retuning of hierarchical PI controllers in a bidirectional off-board fast charger, reducing DC-link settling time from 920 ms to 238 ms and maintaining THD below 1.2%. Read together, these studies indicate that PI-based regulation remains deeply embedded in EV charging systems, but its practical competitiveness increasingly depends on intelligent tuning rather than on conventional manual adjustment alone.

### *C. Group III: Integrated PV–EV Charging and Energy-Management Studies*

The studies reviewed at this level of the literature move beyond isolated charger design and local controller refinement toward a broader system perspective in which EV charging is treated as part of a coordinated PV–EV–grid energy environment. The central question is no longer limited to how the converter is structured or how a given controller is tuned, but rather how multiple subsystems—renewable generation, charging demand, DC-link regulation, battery behavior, and grid interaction—can be coordinated in a way that preserves charging continuity, power quality, and operational efficiency. As shown in Table III, this part of the literature is therefore distinguished by its attempt to elevate EV charging from a

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TABLE II
COMPRESSED COMPARISON OF CONTROL AND OPTIMIZATION STUDIES FOR EV CHARGING SYSTEMS

| Ref. | Year | Controller Family | Optimization / Control Strategy | Principal Control Contribution | Validation Level | Key Reported Improvement | Principal Limitation |
|---|---|---|---|---|---|---|---|
| [15] | 2025 | PID | GWO-optimized PID | Adaptive controller tuning for improved inverter regulation and power quality in a grid-connected PV system | Simulation only | THD reduced to 3.7%; settling time 0.035 s; PF $\approx 1$ | No hardware validation |
| [16] | 2024 | AMP-DPC + ADPC | Adaptive predictive/direct power control | Dual control strategy for bidirectional EV charger power-quality enhancement across G2V/V2G/V2L modes | Experimental prototype | THD reduced from 29.301% to 2.008%; settling time improved to 5–8 ms | No formal optimization framework; limited to 0.5 kW prototype |
| [17] | 2024 | PI | AIChOa-based PI tuning | Hybrid metaheuristic tuning of charger control gains for improved V2G/G2V performance | Simulation only | THD reduced to 4.56%; improved convergence and reduced error metrics | Simulation only; objective function remains simplified |
| [18] | 2025 | Robust nonlinear / sliding-mode-based coordinated control | Optimal coordination methodology | Robust coordinated control for bidirectional BEV chargers under V2G/G2V operation | Simulation + CHIL | Improved tracking and reduced DC-bus error relative to baseline controllers | Optimization settings and statistical validation not fully reported |
| [19] | 2024 | Cascaded PI | WHPSO-based tuning | Metaheuristic optimization of controller parameters for DC-link and battery-current regulation | Simulation + HIL | Overshoot reduced from 27.5% to 3.75%; settling time reduced from 125 ms to 20 ms | No full experimental prototype; HIL includes communication-delay effects |
| [20] | 2025 | Robust PI | SSA-based tuning | Optimization-assisted PI tuning for improved DC-link control, MPPT support, and grid-current quality | Simulation only | THD reduced to 2.50%; settling time improved to 0.015 s | Simulation only; weighting coefficients not fully disclosed |
| [21] | 2025 | Hierarchical PI | ABC-based coordinated tuning | System-wide PI retuning for bidirectional off-board fast-charger control and power-quality improvement | Simulation only | DC-link settling time reduced from 920 ms to 238 ms; THD $< 1.2\%$ | Simulation only; no HIL or experimental validation |

converter-level problem to a system-level coordination problem.

One important direction within this stream is the development of PV-assisted charging platforms that explicitly coordinate charging behavior with grid-side requirements. In [22], a PV-integrated dual-station charging infrastructure is regulated through a dual-SOGI control framework with PI-based coordination, enabling harmonic mitigation, reactive-power compensation, and improved grid performance. The reported grid-current THD of about 1–1.1%, together with a power factor above 0.98 and a reduction in network losses exceeding 25%, indicates that the introduction of PV generation need not weaken grid-side performance if the integration layer is properly coordinated. A related system-oriented contribution appears in [24], where a PV-grid-EV charging system based on a 5-level bidirectional multilevel converter combines PI control, P&O MPPT, and SOGI-PLL synchronization to support PV2EV, PV2G, G2V, and V2G modes within a single coordinated framework. Likewise, [28] extends the architectural scope toward a PV-assisted multi-output charging configuration using PWM-based mode switching, showing that multi-mode routing among PV2V, G2V, and PV2G functions can be achieved within a unified charger structure. Read together, these studies make clear that once renewable integration becomes central, charger design must increasingly be evaluated in terms of coordinated operating modes and grid-facing behavior rather than only in terms of local converter efficiency.

A second major direction is the growing emphasis on supervisory energy management, predictive coordination, and higher-layer decision logic. This tendency is especially visible in [23], [25], [27], and [29]. In [23], the hybrid SWO-MHFAN framework combines optimization and prediction in a grid-connected PV-powered EV charging station, using life cycle cost minimization as the principal management objective. The reported outcomes—THD of 1.04%, power factor of 0.986, and reduced operating cost—show that the charging problem is increasingly being treated as an economic and operational coordination problem rather than purely an electrical regulation problem. In [25], the station manager and MPC framework take a more explicitly real-time approach, minimizing future tracking error to improve PV utilization and charging continuity under changing operating conditions. Study [27] pushes the scope even further by treating V2G, G2V, and V2X as a system-level scheduling problem, where ABCO is used to optimize energy exchange across multiple entities. By contrast, [29] approaches the same system-level challenge through BELBIC-based intelligent control for an autonomous PV-powered charging station, targeting DC-bus regulation and transient improvement without relying on a formal offline optimizer. Although these studies differ substantially in method, they collectively indicate that EV charging research is gradually absorbing the language of energy-management systems: prediction, scheduling, supervisory coordination, and intelligent decision-making are becoming part of the charging problem itself.

Among the reviewed studies, [26] is particularly important because it exposes a cross-layer interaction that is only implicit in much of the surrounding literature. Rather than treating MPPT, grid-side control, and battery-side regulation as adjacent but largely independent functions, it shows that MPPT-induced perturbations can propagate through the common DC-link and materially influence grid-side power quality and transient performance. The comparative results reported in Table III are especially revealing in this regard: before coordinated retuning, the integrated system exhibited charging THD of 4.02%, overshoot of 33.88%, and settling time of 442 ms, whereas after ABC-based retuning these values improved to 1.40%, 26.43%, and 253.8 ms, respectively. The importance of this result extends beyond the numerical gains themselves. It shows that renewable integration cannot be treated simply as the addition of a PV source to an existing charger; rather, it alters the internal dynamics of the entire charging system and therefore requires coordinated retuning across the control

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TABLE III
COMPRESSED COMPARISON OF INTEGRATED PV–EV CHARGING AND ENERGY-MANAGEMENT STUDIES

| Ref. | Year | System Architecture | Management / Control Method | Principal Integration Contribution | Validation Level | Key Reported Improvement | Principal Limitation |
|---|---|---|---|---|---|---|---|
| [22] | 2025 | PV-integrated dual-station EV charging infrastructure | Dual-SOGI coordinated control with PI regulation | Coordinated PV–EV–grid operation for harmonic mitigation, reactive power compensation, and improved grid power quality | Simulation + experimental prototype | Grid current THD $\approx$ 1–1.1%; PF $> 0.98$; network losses reduced $> 25\%$ | No formal optimization framework; controller tuning not explicitly reported |
| [23] | 2025 | Grid-connected PV-powered EVCS with battery storage | Hybrid SWO-MHFAN energy-management framework | Hybrid prediction-and-optimization-based energy management for PV-powered EV charging stations | Simulation only | THD 1.04%; PF 0.986; operating cost 0.124 M$/year | No experimental validation; constraints and statistical analysis not fully reported |
| [24] | 2025 | PV-grid-EV charging system with 5-level bidirectional multilevel converter | PI + P&O MPPT + SOGI-PLL | Renewable-integrated bidirectional charging with multi-mode operation and improved grid power quality | Simulation + HIL | Efficiency 97.1% (G2V), 96.7% (V2G); THD 1.2% / 0.9% | No optimization-based tuning; limited to single-phase low-power configuration |
| [25] | 2024 | Grid-connected PV-EV charging station with station manager and common DC link | Station manager + MPC | Real-time dynamic power management through adaptive on-grid/off-grid mode selection | Semi-experimental (PIL) | MPC IAE 0.4746 vs. 5.434 (PID); improved charging continuity and PV utilization | No V2G capability; implementation cost and complexity remain high |
| [26] | 2026 | Grid-integrated PV-based bidirectional off-board fast charger | Fuzzy VSS-P&O + cascaded PI + ABC retuning | System-level coordination of MPPT, grid-side control, and battery-side regulation in a PV-assisted fast charger | Simulation only | Charging THD improved from 4.02% to 1.40%; DC-link settling time 442 $\rightarrow$ 254 ms | Simulation only; optimization performed offline |
| [27] | 2025 | Conceptual V2G/G2V/V2X energy-exchange system | ABCO-based scheduling / optimization | System-level optimization of bidirectional and multi-entity energy exchange | Simulation only | Charging time 13.09 h vs. 16.8 h baseline; reactive power improved to $-200$ Var | No converter-level modeling; conceptual formulation only |
| [28] | 2024 | PV-assisted grid-connected multi-output EV charger | PWM-based mode switching | Multi-output charging architecture supporting PV2V, G2V, and PV2G functions | Simulation + experimental prototype | Experimental efficiency 90.18%; total output 500 W | No closed-loop supervisory control; THD not reported |
| [29] | 2024 | Autonomous PV-powered EV charging station with storage battery | BELBIC-based intelligent control | Intelligent energy-management-oriented control for DC-bus regulation and charging-station performance enhancement | Simulation only | Overshoot reduced to $\leq$6%; settling time reduced to $\leq$0.08 s | No experimental validation; heuristic tuning only |

hierarchy. In that sense, this study illustrates a broader point that is only partially addressed elsewhere in the literature: the integration layer is not external to converter and control design, but dynamically entangled with both.

## III. LAYERED DEVELOPMENT OF EV CHARGING SYSTEMS: ARCHITECTURE, CONTROL, AND PV-INTEGRATED ENERGY MANAGEMENT

The studies synthesized in Table IV are best understood through a layered analytical lens that captures the transition from hardware realization, to control-oriented refinement, and finally to coordinated PV-integrated energy management. This perspective does not simply classify the literature; it clarifies how the field has matured, where its main technical advances have occurred, and why the next phase of progress increasingly depends on cross-layer design integration.

The grouping logic used in Table IV is based on functional and methodological commonality rather than on identical circuit topology alone. Studies are grouped when they address a shared technical problem, pursue a comparable design intention, perform a similar converter or control function, operate within the same EV-charging application domain, or reveal a common validation limitation. Accordingly, Table IV should be read as a cross-study synthesis matrix that connects technical objectives, topology classes, control methods, validation platforms, and residual gaps, rather than as a strict one-to-one classification of identical architectures.

At the architectural level, the reviewed studies reveal a field driven by negotiated design trade-offs rather than convergence toward a single dominant topology. The first group of contributions is centered on the physical realization of charging capability through distinct converter structures, each tailored to a particular combination of gain requirement, directionality, efficiency target, and battery-voltage range. Studies such as [8], [9], and [10] represent one major direction, namely the use of resonant and isolated bidirectional architectures to combine high efficiency, soft switching, and wider voltage compatibility. These designs are especially important in contexts where the charger is expected to support demanding G2V/V2G operation, FCBEV interfacing, or multi-stage power conditioning with relatively high power-quality requirements. By contrast, studies such as [11], [12], [13], and [14] reflect a different architectural logic, one that emphasizes voltage-range flexibility, interleaving-based ripple mitigation, current-sharing improvement.

What Table IV makes especially visible, however, is that these architectural choices are never neutral. Each one privileges certain forms of performance while leaving other concerns unresolved. Resonant and isolated structures offer strong functional breadth, but frequently remain supported more strongly by simulation or HIL than by full prototype evidence. Interleaved and multiphase structures improve efficiency, ripple behavior, and current sharing over wider load ranges, yet often introduce additional conduction burdens, parameter sensitivity, or scalability concerns. Simplified high-gain structures reduce switch complexity and extend voltage-boosting capability, but commonly do so at the cost of passive-component burden, open-loop reliance, or limited thermal maturity.

This helps explain the second major shift evident in the reviewed corpus, namely the movement from hardware realization toward control-oriented performance engineering. In this layer, the core technical challenge is no longer how to construct the charging interface, but how to govern its behavior under dynamic operating conditions in a way that improves THD, settling behavior, DC-link stability, current tracking, synchronization quality, and bidirectional coordination. Studies such as [15], [16], [17], [18], [19], [20], and [21] make clear that once converter-level functionality becomes sufficiently mature, the decisive performance gains often arise not from new hardware, but from improved control formulation and tuning discipline. A useful distinction emerges here between studies that redesign the control law itself and those that retain familiar

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TABLE IV
COMPRESSED CROSS-STUDY COMPARISON OF TECHNICAL PROBLEMS, DESIGN INTENTIONS, TOPOLOGY CLASSES, CONTROL METHODS, VALIDATION PLATFORMS, AND RESIDUAL GAPS IN THE REVIEWED EV CHARGING LITERATURE

| Ref. | Tech. problem | Design int. | Topol. class | Distinct contrib. | Ctrl. method | Valid. platform | Residual gap |
|---|---|---|---|---|---|---|---|
| [8], [9], [11] [12], [16], [18] [19], [21] | Fast/bi-dir. EV chg. with low THD, ripple suppression, dc-link stability, and robust operation over wide voltage/load ranges | Single-/two-stage charger structures targeting efficient G2V/V2G, soft switching, seamless multimode transfer, and coordinated converter tuning | Resonant AC–DC, VSC+DAB, mod. SEPIC, bi-dir. buck–boost, grid-int. bi-dir. charger | Unified progress in fast charging, soft switching, multimode bi-dir. operation, improved current sharing, and stronger dynamic/grid-interactive performance | ST-SMC, dq/VSC, PI, DPC, MPC, FF, metaheuristic-tuned cascaded PI | Sim. + HIL/CHIL + limited prot. | Full-scale experimental validation is still limited; scalability, practical robustness, and systematic tuning/optimization under real operating uncertainties remain open |
| [10], [14] [15], [17], [20] | High-gain conversion, low ripple/stress, and improved THD, MPPT, and transient performance under variable operating conditions | Compact high-gain converter development and optimization-assisted controller tuning for better regulation and power-quality enhancement | Quadratic boost + DAB + T-type inv.; interleaved boost; UHG single-switch buck–boost; PV + boost + VSI; PV + DC–DC + DC–AC | Combines converter-level gain/stress reduction with AI/metaheuristic-supported tuning to improve regulation quality and dynamic response beyond baseline PI/PID designs | PI, PID, phase-shift PWM, volt.-bal., GWO-, AIChOa-, and SSA-based tuning | Mostly sim. + low-power prot. | Thermal behavior, reproducibility of optimization workflows, and large-scale hardware validation remain insufficiently demonstrated |
| [22], [24], [25] [26], [28], [29] | PV/RES-integrated EV charging under irradiance variation, grid faults, PQ constraints, charging continuity requirements, and multifunctional operation | Grid/PV-assisted charger platforms with adaptive energy routing, harmonic mitigation, reactive support, multiport operation, and post-integration retuning | Dual-station PV–EV–grid, 5L-BMC, common DC-link PV-EVCS, PV-assist. fast charger, multiport charger, autonomous PV-EVCS | Enables PV2V/G2V/V2G-related flexibility, harmonic mitigation, adaptive charging continuity, and explicit coordination between MPPT behavior and charger/grid performance | Dual-SOGI, PI, MPC, fuzzy/intelligent ctrl., PWM switching | Sim. + HIL/PIL + hardware prot. | Several studies still lack formal optimization, supervisory EMS depth, generalized converter-level evidence, or full-scale field validation; some omit comprehensive V2G assessment |
| [23], [27] | System-level EMS, forecasting, load balancing, and coordinated V2G/G2V/V2X power exchange in grid-interactive charging ecosystems | Station- and ecosystem-level scheduling/optimization for multi-source, multi-entity EV energy coordination | Grid-int. PV-EVCS w/ storage; V2G/G2V/V2X system-level framework | Shifts the research focus from converter control toward system-level coordination, predictive management, and exchange optimization across multiple actors | SWO-MHFAN, ABCO scheduling/optimization | Sim. only | Mostly conceptual/system-level; converter-level verification, HIL, and experimental demonstration remain absent |

PI/PID structures but strengthen them through optimization-assisted gain tuning. This pattern is analytically important because it shows that the field has not abandoned classical regulation, but has instead recontextualized it. PI- and PID-based control remain widely used, not because they are sufficient in their default form, but because they remain practically valuable when embedded within more deliberate optimization frameworks. The strongest control-oriented studies therefore do more than report lower THD or shorter settling times. They demonstrate that charger performance is increasingly governed by the quality of the tuning methodology and by the extent to which controller design reflects system coupling rather than isolated loop behavior.

The third and most advanced layer broadens the field further by reframing EV charging as a coordinated energy-management problem embedded within a renewable-aware operational environment. In this part of the literature, the charging station is no longer treated only as a converter or a controlled plant, but as an energy node that must coordinate PV generation, battery dynamics, DC-link behavior, grid exchange, and user-oriented charging objectives. This transition is evident in studies such as [22], [23], [24], [25], [26], [27], [28], and [29].

Within this integration-oriented layer, several distinct trajectories can be identified. One consists of PV-assisted charging architectures that explicitly coordinate charging behavior with grid-support functions, as in [22], [24], and [28]. Another emphasizes supervisory or predictive energy management, as in [23], [25], and [27], where optimization and forecasting are used not simply to regulate electrical variables, but to manage energy allocation, cost, scheduling, and station-level adaptability. A third, and analytically perhaps the most significant, is represented by [26], which shows that PV-side MPPT behavior can propagate through the common DC-link and materially alter grid-side power quality, thereby exposing a genuine cross-layer coupling that is only weakly addressed elsewhere in the literature.

## IV. CROSS-GROUP DISCUSSION, RESEARCH GAPS, AND FUTURE DIRECTIONS

The purpose of this section is to move beyond group-wise description and develop a more integrated interpretation of the literature. Guided by the synthesis presented in Table V, the following discussion identifies the principal cross-group consistencies, highlights the most consequential research gaps, and outlines the future directions most likely to shape the evolution of intelligent EV charging systems. The reviewed literature shows a clear progression from converter- and charger-level design to control-oriented refinement, and, more recently, to PV-coupled and station-level charging operation. However, this evolution has not yet produced a truly unified design paradigm.

The analytical axes in Table V were selected from recurring cross-study concerns observed across the three review layers.

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Unlike Table IV, which compares studies according to technical problem, topology class, control method, and validation platform, Table V organizes the literature according to higher-level synthesis criteria: architecture–control interdependence, objective-function formulation, optimization benchmarking, validation maturity, operational realism, battery-aware control, grid-support compliance, scalability, PV–EV dynamic coupling, reproducibility, computational feasibility, and next-generation intelligent charging. This organization is intended to show where limitations recur across different subfields and where future research requires cross-layer coordination rather than isolated subsystem improvement.

Accordingly, claims regarding validation maturity, benchmarking gaps, and deployment readiness are presented as comparative synthesis findings across the reviewed studies, not as formal bibliometric prevalence statistics or meta-analytic effect-size estimates.

The results of the cross-group synthesis are that any future advances in EV charging research will be less tied to the individual advances made in converter topology or controller tuning, but will focus on the creation of system-level design frameworks. Despite the recent research claiming significant improvements in tracking accuracy, harmonic suppression, transient response, and bidirectional operation, the evaluation of performances is still dominated by limited electrical performance, battery ageing, electrothermal stress, lifecycle cost, and service-level quality is still not sufficiently incorporated in optimization and control targets. Simultaneously, there is still a systemic disconnect between technical feasibility and deployment readiness, as much of the literature still uses simulation, or small scale validation, under deterministic operating conditions that fail to fully model stochastic demand, user behaviour, grid variability and public-charging constraints. The design of battery-aware charging is also underdeveloped, with the ability to regulate voltages and currents well is not a guarantee to the long-term health of a battery and sustainable V2G operation. Likewise, grid-support charging systems and renewable-assisted charging systems have shown great promise, yet their functionality is fundamentally influenced by highly coupled interactions between MPPT and DC-link regulation, storage dynamics, battery control, and grid-side current control. Thus, the second phase of the research must focus on hierarchical co-optimization, transparent benchmarking, realistic multi-scale verification, and scalable architectures that combine charging efficiency, battery-conscious management, renewable coordination, and grid-responsive functionality in a single operational system. In addition to the cross-layer issues emphasized here, several complementary topics deserve more specialized treatment in future work, including wireless and pad-based charging, solid-state-transformer-based charging interfaces, SiC/GaN device-level implementation, and safety-, EMI-, and thermal-packaging considerations. These areas are closely related to deployment readiness but require dedicated analysis beyond the page-limited scope of the present thematic synthesis.

## V. Conclusion

This review examined the development of EV charging systems through a layered analytical framework that connects converter and charger architecture, control and optimization, and integrated PV–EV charging with energy management. Viewed through this structure, the literature shows a clear progression from hardware-oriented design toward increasingly intelligent, coordinated, and system-aware charging platforms. At the architectural layer, the reviewed studies demonstrate that converter design remains driven by trade-offs among efficiency, gain capability, bidirectional operation, ripple suppression, and voltage adaptability, rather than by convergence toward a single dominant topology. At the control layer, the literature shows that once converter functionality becomes sufficiently mature, further performance gains are increasingly achieved through improved control formulation, coordinated regulation, and optimization-assisted tuning. At the integration layer, EV charging is no longer treated as an isolated converter problem, but as part of a broader PV–EV–grid ecosystem in which renewable intermittency, supervisory management, storage interaction, and grid-side constraints must be addressed simultaneously.

The cross-group synthesis reveals, however, that this technical evolution has not yet produced a fully unified design paradigm. Performance assessment remains dominated by narrow electrical metrics, while battery ageing, electrothermal stress, lifecycle cost, and service-level quality are only weakly integrated into optimization and control objectives. In addition, the gap between technical feasibility and deployment readiness remains substantial, since many studies continue to rely on simulation-based or limited-scale validation under deterministic operating conditions. As a result, the robustness and superiority claimed for many proposed methods should be interpreted cautiously when real charging environments are shaped by stochastic demand, user behaviour, communication constraints, grid variability, and infrastructure limitations. The review also shows that battery-aware charging design, grid-code-oriented validation, and large-scale station-level coordination remain insufficiently developed across much of the literature.

Taken together, these findings suggest that the next stage of progress in EV charging research will depend less on incremental improvement within isolated subsystems and more on the development of transferable system-level design principles. Future work should therefore move toward hierarchical co-optimization frameworks that jointly consider converter topology, controller synthesis, battery-aware management, renewable integration, and operational coordination. Greater emphasis is also needed on transparent benchmarking, reproducible optimization protocols, realistic multi-scale validation environments, and integrated evaluation criteria that reflect both technical and deployment-oriented performance. In this sense, the long-term advancement of EV charging systems will depend not simply on making chargers faster or more efficient, but on establishing scalable, adaptive, and grid-responsive

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TABLE V

CROSS-STUDY CRITICAL SYNTHESIS, UNRESOLVED GAPS, IMPLICATIONS, AND FUTURE PRIORITIES IN THE REVIEWED EV CHARGING LITERATURE

| Analyt. axis | Cross-study synthesis | Unres. gap | Crit. impl. | Future priority | Ref. |
|---|---|---|---|---|---|
| Arch.–ctrl. interdep. | Lit. shifts conv./charger design → ctrl. refinement → PV-coupled station op.; recent reviews show chg. arch., storage, renewable interfacing, grid-int. stds., and supervisory data mgmt. are non-separable. | Topology, ctrl. synthesis, supervisory coord. still treated separately. | Conv.-level gains may not hold under DC-link coupling, renewable intermittency, bi-dir. op., and supervisory comm. | Co-design conv. arch., ctrl. tuning, comm. arch., station op. | [8]– [14], [15]–[21], [22]– [29], [30], [3], [4], [31] |
| Obj.-func. | Opt. studies target tracking err., THD, rise/settling time, V/I regulation; reviews add service-op. objs. incl. adequacy, waiting, routing, cost, market-aware scheduling; efficiency, therm. stress, batt. aging, lifecycle rarely unified. | Obj. sets weakly standardized. | "Best optimizer" claims remain methodol. weak under non-equivalent objs. | Standardize multi-criteria objs.: dyn. response, PQ, efficiency, conv. stress, batt.-health, user-service. | [15], [17], [19], [20], [23], [26], [27], [2], [33] |
| Opt. benchmarking | Proposed optimizers typically benchmarked vs. limited baselines under study-specific settings. | Bounds, stopping rules, runs, init., comp. budgets inconsistently reported. | Superiority may reflect benchmark asymmetry, not true alg. advantage. | Enforce equal search space, equal budget, repeated runs, full hyperparameter disclosure. | [15], [17], [19], [20], [23], [26], [27] |
| Valid. maturity | Corpus spans simulation-only, HIL, and fewer prototypes. | Realistic full-scale validation remains limited. | Technical feas. exceeds deployment readiness. | Expand HIL, RTDS, prototype validation for integrated PV-assisted bi-dir. platforms. | [8], [12], [13], [16], [18], [19], [21], [22], [24], [25], [28] |
| Op. realism + uncert. | Most studies use deterministic scenarios; reviews show user choice, chg. location, stochastic demand, waiting, pricing response, real-grid var. remain underrepresented. | EV arrivals, user behavior, SoC diversity, comm. delays, broader grid uncert. weakly modeled. | Robustness may be overstated. | Use uncert. ensembles, real chg. datasets, mission-oriented profiles. | [15]– [21], [22]–[26], [28], [29], [1], [2], [32], [33], [6] |
| Batt.-aware ctrl. | Batt. regulation is explicit in many bi-dir./PV-assisted studies; yet reviews confirm degrad., electrotherm. dyn., chg.-strategy effects, V2G wear remain weakly integrated; newest V2G review reinforces this. | Batt. degrad. + electrotherm. + lifecycle-aware opt. remain under-modeled. | Fast response alone is insufficient absent long-term batt. sustainability. | Embed aging, electrotherm., lifecycle-cost in low-level ctrl. + upper-layer EMS. | [16], [18], [19], [22], [23], [25], [26], [27], [29], [34], [5], [7] |
| Grid-support/compliance | Growing subset addresses PF correction, harmonic mitigation, bi-dir. grid support, market-aware coord.; reviews stress smart chg., V1G/V2G, ancillary services, renewable balancing, grid-code compatibility; new V2G review reinforces peak shaving, freq. regulation, renewable support, grid stabilization. | Grid-support often shown at limited scale, not broad code-oriented tests. | Technical promise evident; utility-grade robustness still under-demonstrated. | Advance grid-code-compliant V2G/V2X incl. weak-grid behavior, FRT, comm. interoperab., reactive-power support. | [16], [18], [19], [21], [22], [24], [25], [26], [28], [1], [2], [3], [4], [5], [6], [7] |
| Scalability | Lit. remains largely charger-/conv.-centric; reviews show infrastructure planning, station capacity, public-chg. op., routing, multi-user service design are context-dependent. | Multi-charger coord., shared DC bus, fleet/aggregator op. underdeveloped. | Isolated-charger results are weak proxies for real fast-chg. hubs. | Shift to station-level, multi-charger, aggregator-aware models. | [22], [23], [25], [26], [28], [32], [30], [33] |
| PV–EV dyn. | PV int. changes the ctrl. problem itself; MPPT perturbations may propagate via common DC-link to grid-side PQ; reviews further show renewable-enabled smart chg. requires coordinated treatment of power electronics, storage, chg. modes, grid interaction; solar-powered review frames PV-grid, PV-standalone, hybrid chg. arch. as coupled techno-economic systems. | MPPT–DC-link–batt.–grid coupling remains insufficiently resolved. | Renewable int. is a dyn. coupling problem, not merely energy supply. | Hierarchical coord. among MPPT, DC-link regulation, batt.-side loops, grid-side current ctrl., solar-storage mgmt. | [22], [25], [26], [28], [29], [30], [4], [31] |
| Reprod./transp. | Reporting remains uneven: runs, stats., bounds, optimizer settings, infrastructure assumptions, scenario comparability often omitted; reviews also note limited real-world datasets, inconsistent assumptions, ambiguous definitions. | Incomplete reporting weakens reprod. + cumulative comparison. | Methodol. maturation slows. | Report bounds, runs, convergence, data provenance, comp. settings, hardware constraints. | [15], [16], [17], [19], [22], [23], [24], [26], [27], [2], [32], [6] |
| Comp. feas./deploy. | Opt.-assisted methods often show strong offline performance; only limited subset addresses real-time/embedded feas.; reviews further stress scheduling, IoE-based coord., pricing responsiveness, secure comm., field constraints, and infrastructure limits remain undervalidated. | Real-time comp. burden, online retuning, memory, embedded trade-offs inconsistently assessed. | Strong offline results may still be field-impractical. | Evaluate comp. efficiency + deploy. alongside ctrl. quality. | [16], [18], [22], [25], [26], [27], [29], [2], [3], [33], [6], [7] |
| Next-gen. intelligent chg. | Lit. moves from isolated conv. enhancement toward integrated, adaptive, renewable-aware, user-aware, market-aware, data-driven chg. ecosystems; reviews reinforce service op., V2G, renewable coord., IoE-enabled ctrl., grid-impact mitigation, solar-powered arch., intelligent infrastructure planning. | Unified framework linking conv. topology, ctrl. synthesis, renewable int., batt. intelligence, service op., station coord. remains missing. | Lack of such a framework limits transferable design principles. | Advance self-adaptive, hierarchical, multi-objective, secure, digitally integrated EV chg. platforms. | [8]– [29], [1], [2], [32], [34], [30], [3], [33], [4], [5], [6], [31], [7] |

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charging ecosystems capable of operating reliably under real-world conditions.

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